RESEARCH ARTICLE

Global shifts in the geopolitics of chemical discovery: China's rise and the decline of US influence

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ABSTRACT

At the turn of the 21st century, China has achieved a spectacular surge in the scientific arena. However, its influence on the material core of chemistry, encoded in the chemical space of discovered substances, remains unexplored. Here we show, by analyzing the chemical space between 1996 and 2022, that its expansion has been dominated by China ever since 2013. Chinese dominance occurs at different levels, from organic to rare-earth chemistry, but is less dramatic in organometallic chemistry. We also found that Chinese dominance is mainly the product of the country's own efforts, rather than the result of international collaboration. China's surge mainly comes at the expense of the contribution of the United States. Interestingly, the US share of the chemical space is more dependent on international collaboration, which mainly occurs with China. We also observe the emergent role of India. We believe these results provide a contemporary account of the geopolitics of the chemical space, which may constitute the basis for future national and international science policies, as well as research and development agendas.

1. INTRODUCTION

China's remarkable rise in the scientific and technological arena of the 21st century is well documented, along with the factors contributing to this growth (National Science Board, 2018; Dhand et al., 2024; Woolston, 2023; Zimmermann, 2025). However, limited attention has been given to China's influence on the recent development of chemistry—the most prolific natural science in terms of publications and arguably the most significant in its economic and societal impact (Jost & Restrepo, 2022).

Often, studies analyzing the rise of China take a broader scientific and technological scope (National Science Board, 2018; Woolston, 2023) and do not provide a detailed picture of China's role in the current dynamics of chemistry. An exception to this trend is the recent report by the American Association for the Advancement of Science (AAAS), which only in its electronic dashboard—and not in the printed document—provides information on the dynamics of chemistry publications (journals and patents) for the last decade (Zimmermann, 2025).

The AAAS dashboard reveals that highly cited chemistry journal publications from the United States are declining, whereas those from China are rapidly increasing (Figure S1a in

an open access journal

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the Supplementary material; Zimmermann, 2025). Likewise, Chinese patent applications in semiconductors have surged, positioning the country as the leading patent applicant, while the United States ranks third after Japan (Figure S1b; Zimmermann, 2025). Nonetheless, in the field of pharmaceutical patents, the United States remains as the leading applicant, followed by China, whose patent applications are the most rapidly growing (Figure S1c) (Zimmermann, 2025). Despite these important results, they are based entirely on the "paper" and the "patent" as the objects used to measure the scientific and technological output of chemistry. We argue that there is a further and essential aspect of chemical practice that has so far remained unexplored: the discovery of substances, which lies at the core of chemistry (Schummer, 1998). Together, the discovery of chemicals and their associated chemical reactions constitute what is commonly referred to as the *chemical space* (CS)—a conceptual framework encompassing the totality of known substances and their chemical reactions (Llanos, Leal et al., 2019; Restrepo, 2022).

Here we quantify China's contribution to the CS and contrast it with contributions of other leading countries, especially the United States. Likewise, we quantify the role of international collaboration in the expansion of the CS and three of its important subspaces, namely the organic, organometallic, and rare-earth regions of the CS.

2. RESULTS

The study is based on information retrieved from Reaxys©[1], Dimensions, and OpenAlex databases for the time period 1996–2022 (Figure S2), which shows China's predominant role in the CS as it covers 41% of the current CS, dwarfing the 11% of the United States (Figure 1a, Figure S3). While China grew exponentially during this period (see the Supplementary material), US participation grew modestly until 2007 and afterwards dropped rapidly (Figure 1a). In 2013, China became the overall leading contributor. The only other country whose share was systematically increasing is India, but its growth is modest compared with that of China (Figure 1a). The share of the CS for 155 countries can be visualized in the online dashboard (https://countrywise-chemical-space.com/CS-Explorer-App/) associated with this paper.

2.1. National and Collaborative Efforts to Expand the Chemical Space

The pivotal role of international collaboration in China's scientific and technological rise in the 21st century—particularly with the United States—has been well documented (Hu, 2020; Van Noorden, 2022; Zhou & Glänzel, 2010) Our analysis reveals that such collaboration is a dominant feature within the CS (Figure 1b). However, its overall contribution remains relatively modest, accounting for less than 2% of all chemical discoveries (Figure 1b). The volume of China–US collaborative output increased steadily over time, peaking in 2017, but has shown significant volatility since then (Figure 1b). Further trends in international collaboration contributing to the expansion of the CS can be explored through the dashboard associated with this paper.

Notwithstanding the China–US collaboration, over 90% of China's contribution to the CS was achieved by domestic teams operating independently. In contrast, the United States has seen a decline in solo contributions, which exceeded 95% between 1996 and 2006 but sharply decreased to less than 80% of its participation in the CS expansion (Figure 1c). The drop in US solo contributions has been counterbalanced by an increase in international

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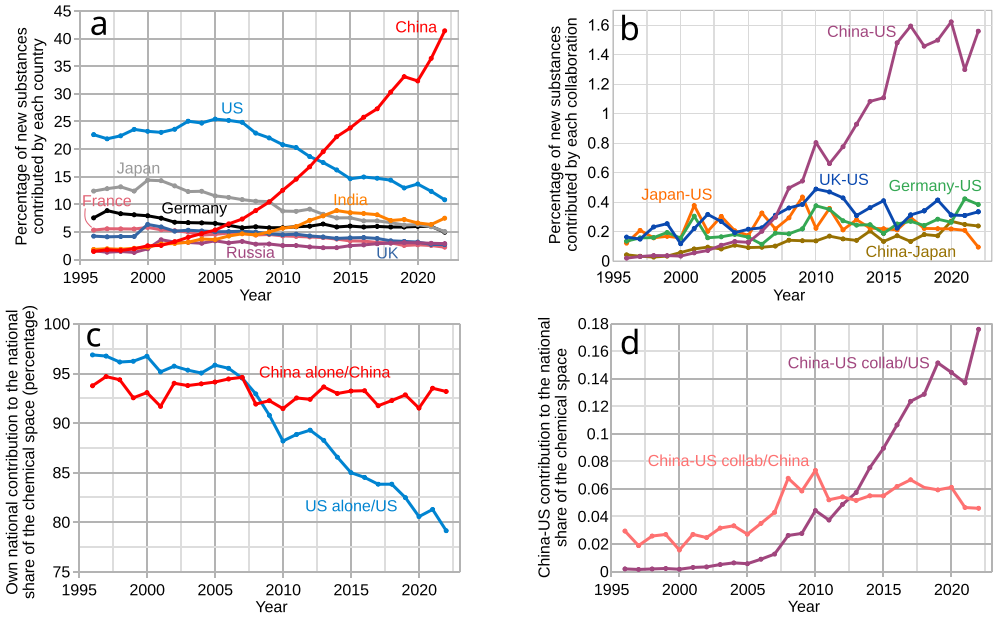


Figure 1. Countrywise expansion of the chemical space (CS). (a) Country participation in the growth of the CS. Only the top eight countries are shown (further information in the Supplementary material and in the dashboard accompanying this paper). (b) Five of the most relevant international collaborations in the CS (see Supplementary material). All other international collaborations can be visualized in the dashboard. (c) Percentage of new substances with participation of country *c* that are reported in papers with no international collaboration (see Supplementary material). (d) Percentage of new substances with participation of country *c* (*c* is either China or the United States) resulting from China–US collaboration. Note that China–US collaboration may include collaborations with other countries (see Supplementary material).

collaboration (Figure 1b), particularly with China (Figure 1b, d). Meanwhile, the United States has maintained a relatively stable partnership with European countries (Figure 1b).

As intranational collaboration is more likely for larger countries than for smaller ones (Schubert & Braun, 1990), we also corrected for country size in the collaboration between China and the United States. The results (Figure S4) indicate that country size, as measured by publication volume, is not the driving factor of collaboration. Therefore, Chinese preference for domestic rather than international collaboration is not a statistical effect but appears to be driven by a national or cultural way of practicing chemistry.

2.2. 2008 and 2020 Local and Global Drops on the Expanding Chemical Space

We observe effects, with a certain delay, of the 2007/2008 global financial crisis for the US contribution to the CS and of the COVID-19 pandemic for China (Figure 1a). In the long run, the 2007/2008 crisis did not reduce the expansion of the CS, whereas the pandemic did (Figure 2a, Figure S5). Although China experienced a drop in researchers by 2007/2008 (Figure S8 and the dashboard in Zimmermann (2025)), this did not affect its increasing contributions to the CS, which counterbalanced the declining US contributions (Figure 1a). The pandemic and the rapid lockdowns in China (Jia, 2024) seem to have reduced the contributions of the Asian country and, therefore, of the whole CS (Figure 2a). In contrast, the more relaxed US lockdowns (Pilkington & McCarthy, 2020) did not greatly affect the US contributions to the CS (Figure 1a).

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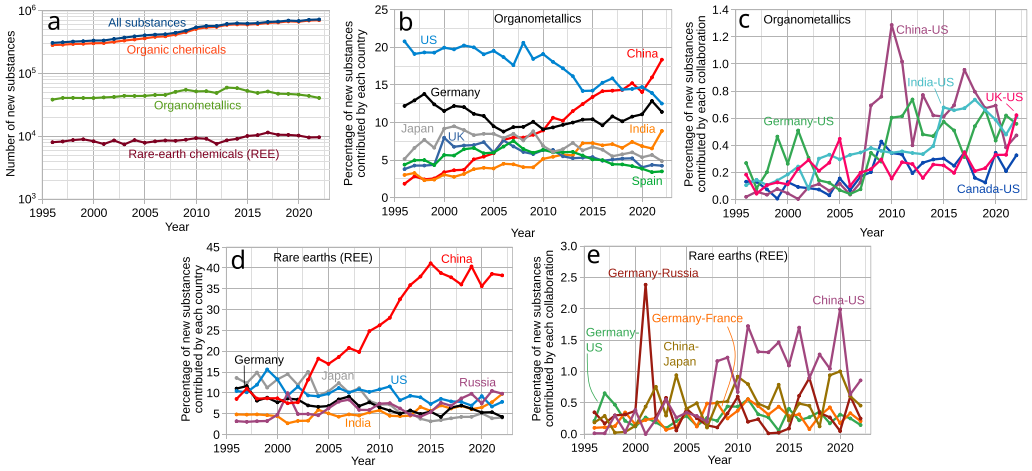


Figure 2. Countries' share of the expansion of subregions of the chemical space (CS). (a) Recent expansion of the CS and of three of its subspaces (see Supplementary material, dashboard). CS grows at an annual rate of 3.8%, calculated by linear regression of the "All substances" curve (R^2 = 0.9724). (b) Countries' participation in the expansion of the organometallic CS and its (c) top 5 international collaborations. (d) Countries' participation in the expansion of the REE CS and its (e) top 5 international collaborations. The criteria for the selection of the substances belonging into these spaces are found in the Supplementary material. Temporal trends for each of the 155 countries analyzed in this paper, along with their collaborations, are found in the dashboard accompanying this paper.

2.3. Chemical Space, its Subspaces and National Contributions to Their Expansions

We divided the CS into its organic, organometallic, and rare-earth (REE) subspaces (Figure 2a, Supplementary material), as they capture large and traditional areas of research in chemistry, as well as novel fields that are central to current geopolitical developments. The organic subspace encompasses all carbon-containing compounds. The organometallic subspace includes compounds containing carbon and at least a metal. The rare-earth subspace comprises all substances containing any of the 17 rare-earth elements, spanning yttrium (Y) to lutetium (Lu) in the periodic system.

We found that the exponential expansion of the CS, as observed since 1800 (Llanos et al., 2019), is largely dominated by the discovery of organic chemicals (Figure 2a). In contrast to the swift growth of the organic subspace, the organometallic and REE subspaces exhibit a distinct trend (Figure 2a). The former slightly increased its expansion until 2013 and ever since has slowed down its unfolding. The REE chemical space, in turn, grows slowly. Shares of these subspaces for 155 countries, as well as the role of international collaboration in their expansion, can be visualized in the dashboard associated with this publication.

2.3.1. Organic space

As organic chemistry encompasses the majority of the CS (Llanos et al., 2019), the overall growth of CS mirrors the expansion of organic chemistry, with China currently contributing about 40% of the organic CS (Figure S7, dashboard) and China–US collaboration accounting for about 2% of this subspace (Figures S4b, S8, dashboard). Likewise, the 2007/2008 crisis did not affect the organic CS, while the 2020 crisis did (Figure 2a, Figure S7).

2.3.2. Organometallic space

Unlike the organic CS, the Chinese leading role in the organometallic CS is very recent and much less pronounced (Figure 2b). While the contributions of the United States and Japan decreased rapidly, not only China, but also India and Germany increased their share (Figure 2b, dashboard). At the collaborative level, China–US partnership remarkably

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increased in 2008 and kept growing until 2010, when it began to oscillate (Figure 2c, Figure S4c, dashboard). The pandemic marked the change of dominant country in the expansion of the organometallic space, with the United States stepping aside and giving way to China as the leading discoverer of organometallics (Figure 2b). The pandemic also brought a drop in China–US collaboration (Figure 2c, Figure S4c, dashboard), which suggests that China's postpandemic rise was driven by its own national efforts.

2.3.3. REE space

We found that the REE space grows slowly (Figure 2a, dashboard). While earlier the shares were distributed among several countries, China had already become dominant by 2004, with a share of the REE space of about 40%, while most of the top contributors slowed down their discoveries (Figure 2d, dashboard). The drop is particularly dramatic for Japan. India, in contrast, grew from 5% in 1996 to 10% today. Russia also grew, but less steadily. Currently, however, China's contribution is four times that of the second-largest producers of REE compounds: India and Russia (Figure 2d).

China–US collaboration has grown from less than 1% of the share of the REE space before 2008 to 2% in 2020. However, over the following two years, collaboration declined, and it currently represents 0.86% of the REE CS (Figure 2e, Figure S4d, dashboard).

Despite the geopolitical dynamics of the CS, the chemical nature of the CS has remained untouched, as the chemical composition of substances of each one of the three analyzed chemical subspaces has not been affected (Figure S9).

3. DISCUSSION

Chemical space (CS)—the collective body of known chemicals and their reactions—has grown steadily by around 4% per year for over 200 years, largely unaffected by changes in social, economic, or geopolitical conditions (Jost & Restrepo, 2022; Llanos et al., 2019). While France, Germany, and later the United States were historically the main hubs of chemical research (Bensaude-Vincent & Simon, 2008), a significant shift is now taking place, with China emerging as a new center of innovation. This shift has not altered the overall rate of expansion of CS (Figure 2a) but rather has reconfigured its contributors. Thus, the increasing contributions of China to the CS come at the expense of other countries; particularly of the United States, whose share is dramatically decreasing. This trend has also been observed in other scientific and technological domains (Zimmermann, 2025).

China's growth in the CS has been driven primarily by its own domestic efforts (Figure 1c) rather than international partnerships, especially with the United States. In contrast, research output from the United States in the CS is becoming increasingly dependent on collaboration with Chinese teams. This trend challenges the assumptions behind recent US policies, such as the "China Initiative" (2018–2022) (Van Noorden, 2022), which claimed that China's scientific progress was mainly a result of cooperation with the United States (Ball, 2024). The reliance of US science and technology on collaborations with the Asian country in particular, has also been documented in other scientific disciplines (Dhand et al., 2024; Lee & Haupt, 2020).

The unfolding of the China–US collaboration has shown notable shifts influenced by geopolitical developments. This partnership grew steadily, peaking around 2017 (Figure 1b), coinciding with a rise in bilateral agreements during the Obama administration (Li, 2016). However, since 2017, collaboration has become more volatile (Figure 1b) and the collaborative affinity between the two countries has declined (Figure S4), likely due to political tensions such as the China Initiative and US-imposed trade tariffs introduced under President Trump

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and continued thereafter (Office of the United States Trade Representative, 2018; Tullo, 2025). These changes have been accompanied by a mutual decline in public and governmental sentiment between the two countries (Xie, Yang et al., 2024).

Still at the level of international collaboration, we foresee potential shifts in the mild but stable partnership between the United States and European nations (Figure 1b). These changes are likely to be driven by recent developments in US immigration policy and reductions in federal research funding (Garisto, 2025; Tollefson, 2025), and, in turn, by the measures adopted by Europe to attract scientific talent from the United States (Zainzinger, 2025).

While China and the United States have experienced the most significant shifts in the global landscape, most other nations have seen relatively stable positions (Figure 1a, dashboard). India, however, has gradually increased its share, whereas Japan's contribution is slowly declining, though less dramatically (Figure 1a). If India backs this upward trend with strong economic investment and support for research and development, it could harness its large population to expand its presence in the CS, potentially leading to broader scientific and economic gains.

China has established a dominant presence across various areas of the CS, including organic, organometallic, and rare earth chemistry (REE). By 2013, it had become the leading contributor of new organic compounds, accounting for 40% of discoveries—far surpassing the United States (11%) (Figure S7, dashboard). This reflects not only the breadth of organic chemistry but also China's cultural ties to traditional medicine, economic reforms, and the growth of its pharmaceutical sector (Ball, 2017; Chen, Xue et al., 2019; Zheng & Li, 2016). In particular, the economic liberalization of the 1980s and 1990s created a fertile environment for multinational pharmaceutical companies to establish a foothold in China. By the 2000s, this was followed by robust domestic research and development initiatives, further bolstered by regulatory reforms that accelerated clinical trial processes and drug approvals (Chen et al., 2019). These developments point to a broader shift in pharmaceutical innovation towards Asia. Recent analyses underscore that this transition is already in motion (Figure S1c; Wainer, 2025).

China's rise as a major contributor to organometallic chemistry is relatively recent, only surpassing other nations in 2020 by a narrow margin. It now accounts for 18% of new organometallic compound reports, ahead of the United States at 13%, with Germany and India also showing growth (Figure 2b, dashboard). This delayed prominence stems from China's earlier emphasis on pharmaceuticals and agriculture, which prioritized traditional organic chemistry (Johnson, Furukawa et al., 2023). Although global interest in organometallics surged in the 1950s following the synthesis of ferrocene (Astruc, 2007), China's engagement gained momentum later, bolstered by institutional support, including the founding of a dedicated laboratory in 1989 (State Key Laboratory of Organometallic Chemistry, 2025). With continued investment, China is poised to lead future advances in catalysis, synthesis, material sciences, and related fields—domains in which organometallics play a pivotal role (Deng & You, 2016).

Rare earth elements have become increasingly significant in technological, economic, and political terms (Balaram, 2019), with China playing a dominant role in their extraction and trade (Ball, 2024). This has led to international tensions (Kullik, 2019) and the establishment of research programs aimed at fostering innovation in REE chemistry (Cuellar, 2021; European Commission, 2020; Hurst, 2010). Since the early 2000s, Chinese research institutions have taken the lead in synthesizing new REE-based materials, marking a shift from earlier decades when scientific innovation was primarily driven by the United States and Japan (Figure 2d, dashboard). Today, China not only controls the extraction and trade of REEs but also leads the exploration of their CS (Figure 2d, dashboard). This transformation reflects

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China's strategic focus on REE scientific advancement, which began with an emphasis on REE extraction and trade in the 1990s (Wang & Dou, 1996), followed by efforts to foster basic research and technological innovation (Shen, Moomy, & Eggert, 2020). China's dominance in REE science carries profound implications for sectors reliant on REE, including computing, telecommunications, renewable energy, and defense. This has led to geopolitical tensions, particularly between China and the United States. The United States has launched policy efforts to contain China's technological progress in areas such as artificial intelligence and to diversify global REE supply chains beyond China's reach (Freifeld, 2025; McCormick, Hodgson, & Stacey, 2025). In turn, China has restricted exports of seven strategically significant REEs, which are predominantly mined and processed within its borders (Jackson, Lv et al., 2025). Chinese dominance of the REE CS is also evident at the patent level. For instance, the country is the leading applicant for semiconductor-related patents (Figure S1b), in which REEs play an important role (Gorelenok, Kamanin, & Shmidt, 2003).

The geopolitics of CS is shifting, with China set to gain prominence as the United States shows signs of retreat (Zimmermann, 2025). Although the United States has traditionally invested more of its GDP in research and development (Figure S10 and the dashboard in Zimmermann, 2025), recent policy changes—such as budget cuts, hiring freezes, and shifts in funding priorities—have created a challenging environment for scientists, particularly those early in their careers (Garisto, 2025; Kozlov, 2025; Milman, 2025). In contrast, China is actively bolstering its scientific workforce and infrastructure (Kennedy, 2025; Peng, 2025; State Council of the People's Republic of China, 2025). Since 2005, it has consistently employed more researchers than the United States (Figure S6 and dashboard in Zimmermann, 2025) and faces lower public debt (Figure S11). With academic research playing a key role in patent innovation (Veugelers & Wang, 2019), China's influence in patented chemicals is expected to grow rapidly, a trend that has already begun to emerge (Figure S1; Dhand et al., 2024; Rammer, Frietsch et al., 2024; Zimmermann, 2025). Despite these changes, the long-term resilience of CS (Llanos et al., 2019) gives hope that global scientific progress will endure.

4. CONCLUSIONS

Over the past 2 centuries, the discovery of chemicals that constitute an essential part of the chemical space (CS) has advanced at a remarkably stable rate, despite profound geopolitical disruptions. This study reveals that the global center of chemical innovation is shifting from the United States to China, without altering the overall pace of growth in CS. China's rapid rise now positions it as the foremost contributor to the discovery of substances. Contrary to claims that China's scientific progress is dependent on US collaboration, we find the inverse: The US increasingly relies on international, especially Chinese, partnerships. These patterns reflect broader geopolitical currents that may reshape global scientific leadership, highlighting China's growing strategic dominance in fields critical to pharmaceuticals, materials science, energy technologies, and defense.

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AUTHOR CONTRIBUTIONS

Marisol Bermúdez-Montaña: Formal analysis, Software, Validation. Angel Garcia-Chung: Formal analysis, Validation. Guillermo Restrepo: Conceptualization, Validation, Writing—

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COMPETING INTERESTS

The authors have no competing interests.

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DATA AVAILABILITY

Chemistry data used in this work is property of RELX Intellectual Properties SA. Bibliographic data (in part) is the property of Digital Science.

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